```
!python --version
```

```
Python 3.12.13
```

```
import os
import warnings
from dataclasses import dataclass

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import display,Markdown
from scipy.optimize import differential_evolution, minimize

warnings.filterwarnings("ignore")
np.random.seed(42)
```

Input param

```
S0 = 232.90
RISK_FREE_RATE = 0.015
TRADING_DAYS = 250
OPTION_FILE = "/content/MScFE 622_Stochastic Modeling_GWP1_Option data.xlsx"
OUTPUT_DIR = "/content/output_dir"
os.makedirs(OUTPUT_DIR, exist_ok=True)
```

```
# Fixed Gauss-Legendre quadrature grids for fast pricing.
LEWIS_NODES, LEWIS_LIMIT = 64, 100.0
x_l, w_l = np.polynomial.legendre.leggauss(LEWIS_NODES)
U_LEWIS = 0.5 * LEWIS_LIMIT * (x_l + 1.0)
W_LEWIS = 0.5 * LEWIS_LIMIT * w_l

CM_NODES, CM_LIMIT = 80, 120.0
x_cm, w_cm = np.polynomial.legendre.leggauss(CM_NODES)
U_CM = 0.5 * CM_LIMIT * (x_cm + 1.0)
W_CM = 0.5 * CM_LIMIT * w_cm
```

```
# load option_data
option_data = pd.read_excel(OPTION_FILE)
option_data.columns = [str(col).strip().lower().replace(" ", "_") for col in option_data.columns]
display(option_data.head())
```

	days_to_maturity	strike	price	type
0	15	227.5	10.52	C
1	15	230.0	10.05	C
2	15	232.5	7.75	C
3	15	235.0	6.01	C
4	15	237.5	4.75	C

```
option_data = option_data.rename(columns={"days_to_maturity": "days","type":"option_type"})
option_data["option_type"] = option_data["option_type"].astype(str).str.upper().str.strip()
option_data["T"] = option_data["days"] / TRADING_DAYS
display(option_data.head())
```

	days	strike	price	option_type	T
0	15	227.5	10.52	C	0.06
1	15	230.0	10.05	C	0.06
2	15	232.5	7.75	C	0.06
3	15	235.0	6.01	C	0.06
4	15	237.5	4.75	C	0.06

Start coding or generate with AI.

```
# helper function
def get_maturity_slice(df,days):
```

```
  sub = df[df["days"] == days]
  strikes = np.array(sorted(sub["strike"].unique()),dtype=float)
  calls = np.array([sub[(sub["strike"]==k)& (sub["option_type"]=="C")]["price"].iloc[0] for k in strikes])
  puts = np.array([sub[(sub["strike"]==k)& (sub["option_type"]=="P")]["price"].iloc[0] for k in strikes])
  return strikes, calls, puts, days/TRADING_DAYS
```

```
def parity_diagnostic(strikes, calls, puts, T):
  rhs = S0-strikes*np.exp(-RISK_FREE_RATE*T)
  dev = calls-puts-rhs
  lb_mse = np.sum(0.5*dev**2)/(2*len(strikes))
  return pd.DataFrame({
      "strike":strikes,
      "call_minus_put":calls-puts,
      "S0_minus_discounted_K":rhs,
      "parity_deviation":dev,}),lb_mse

def calls_to_puts_by_parity(calls,strikes,T):
  return np.maximum(calls-S0+strikes*np.exp(-RISK_FREE_RATE*T),0.0)

def make_fit_table(strikes,mc,mp,model_calls,T):
  model_puts = calls_to_puts_by_parity(model_calls,strikes,T)
  return pd.DataFrame({
      "Strike":strikes,
      "Market Call":mc,"Model Call":model_calls,"Call Error":model_calls-mc,
      "Market Put":mp,"Model Put":model_puts,"Put Error":model_puts-mp,
  })

def plot_fit(table,title,filename):
  plt.figure(figsize=(9,6))
  plt.scatter(table["Strike"],table["Market Call"],label="Market calls",marker="o")
  plt.plot(table["Strike"],table["Model Call"],label="Model Call",marker="x",linestyle="--")
  plt.scatter(table["Strike"],table["Market Put"],label="Market puts",marker="o")
  plt.plot(table["Strike"],table["Model Put"],label="Model Put",marker="x",linestyle="--")
  plt.xlabel("Strike");plt.ylabel("Option Price");plt.title(title)
  plt.legend();plt.grid(alpha=0.3);plt.tight_layout()
  path = os.path.join(OUTPUT_DIR,filename)
  plt.savefig(path,dpi=300);plt.show()
  return path


def plot_errors(table,title,filename):
  plt.figure(figsize=(9,6))
  plt.axhline(0,linestyle="--",linewidth=1)
  plt.scatter(table["Strike"],table["Call Error"],label="Call Error")
  plt.scatter(table["Strike"],table["Put Error"],label="Put Error")
  plt.xlabel("Strike");plt.ylabel("Error");plt.title(title)
  plt.legend();plt.grid(alpha=0.3);plt.tight_layout()
  path = os.path.join(OUTPUT_DIR,filename)
  plt.savefig(path,dpi=300);plt.show()
  return path
#
def plot_errors(table,title,filename):
  plt.figure(figsize=(9,6))
  plt.axhline(0,linestyle="--",linewidth=1)
  plt.scatter(table["Strike"],table["Call Error"],label="Call Error")
  plt.scatter(table["Strike"],table["Put Error"],label="Put Error")
  plt.xlabel("Strike");plt.ylabel("Error");plt.title(title)
  plt.legend();plt.grid(alpha=0.3);plt.tight_layout()
  path = os.path.join(OUTPUT_DIR,filename)
  plt.savefig(path,dpi=300);plt.show()
  return path

def params_to_dataframe(params,mse,model_name,fields):
  rows = [(f,getattr(params,f)) for f in fields]+[("MSE",mse),("RMSE",np.sqrt(mse))]
  out = pd.DataFrame(rows,columns=["parameter","value"])
  out.insert(0,"model",model_name)
  return out
```

```
# Heston Lewis pricing model
```

```
@dataclass
class HestonParams:
  kappa:float
  theta:float
  sigma_v:float
  rho: float
  v0:float
```

```
def heston_return_cf(u,T,r,p):
  u = np.asarray(u,dtype=complex)
  c1=p.kappa*p.theta

  c2 = np.sqrt((p.rho*p.sigma_v*u*1j-p.kappa)**2 - p.sigma_v**2*(-u*1j-u**2))
  c2 = np.where(np.real(c2) < 0, -c2, c2)
  c2 = -c2
  c3= (p.kappa-p.rho*p.sigma_v*u*1j+c2)/(p.kappa-p.rho*p.sigma_v*u*1j-c2)
  e = np.exp(c2*T)

  eps = 1e-12

  H1 = (r*u*1j*T+(c1/p.sigma_v**2)*((p.kappa-p.rho*p.sigma_v*u*1j+c2)*T-2.0*np.log((1.0-c3*e+eps)/(1.0-c3+eps))))
  H2= ((p.kappa-p.rho*p.sigma_v*u*1j+c2)/p.sigma_v**2)*((1.0-e)/(1.0-c3*e+eps))

  return np.exp(H1+H2*p.v0)


def heston_lewis_call_prices(strikes,T,p):
  u=U_LEWIS
  cf=heston_return_cf(u-0.5j,T,RISK_FREE_RATE,p)

  expo = np.exp(1j*u[:,None]*np.log(S0/strikes[None,:]))

  integrand = np.real(expo*cf[:,None])/(u[:,None]**2+0.25)
  integral = W_LEWIS@integrand

  calls = S0-(np.sqrt(S0*strikes)*np.exp(-RISK_FREE_RATE*T)/np.pi)*integral
  return np.maximum(calls,0.0)


def heston_residuals(x,K,MC,MP,T):
  p = HestonParams(*x)

  if 2*p.kappa*p.theta <= p.sigma_v**2:
    return np.ones(2*len(K))*1e5

  try:
    c = heston_lewis_call_prices(K,T,p)
    pu = calls_to_puts_by_parity(c,K,T)
    r = np.r_[c-MC,pu-MP]
    return r if np.all(np.isfinite(r)) else np.ones(2*len(K))*1e5
  except Exception:
    return np.ones(2*len(K))*1e5


def heston_mse(x,K,MC,MP,T):
  return float(np.mean(heston_residuals(x,K,MC,MP,T)**2))


def calibrate_heston_lewis(K,MC,MP,T):
  bounds = [(0.01,20.0),(1e-6,2.0),(0.01,5.0),(-0.999,0.999),(1e-6,2.0)]
  obj = lambda x: heston_mse(x,K,MC,MP,T)

  g = differential_evolution(obj,bounds,seed=777,maxiter=80,popsize=12,tol=1e-8,polish=False,updating="immediate",workers=1
  l = minimize(obj,g.x,method="L-BFGS-B",bounds=bounds,options={"maxiter":1000,"ftol":1e-14})

  return HestonParams(*l.x),obj(l.x)
```

Start coding or generate with AI.

Step 1(a): slice 15 day options & parity check

```
K15,C15,P15,T15=get_maturity_slice(option_data,15)
parity15,party15_lb=parity_diagnostic(K15,C15,P15,T15)
display(Markdown(f"**15-day slice** {len(K15)} strikes,T= {T15:.5f} years"))

display(parity15)

display(Markdown(f"Parity-constrained lower bounds MSE: **{party15_lb:0.6f}**"))
```

15-day slice 5 strikes,T= 0.06000 years

	strike	call_minus_put	S0_minus_discounted_K	parity_deviation
0	227.5	6.20	5.604658	0.595342
1	230.0	4.85	3.106907	1.743093
2	232.5	1.30	0.609156	0.690844
3	235.0	-1.55	-1.888595	0.338595
4	237.5	-4.03	-4.386346	0.356346

Parity-constrained lower bounds MSE: **0.205585**

Start coding or generate with AI.

Step 1(a)- Calibrate Heston-Lewis & report parameters

```
heston_params,heston_mse_val = calibrate_heston_lewis(K15,C15,P15,T15)
heston_param_table = params_to_dataframe(heston_params,heston_mse_val,"Heston-Lewis",["kappa","theta","sigma_v","rho","v0"

heston_param_table.to_csv(os.path.join(OUTPUT_DIR,"step1a_heston__lewis_params.csv"),index=False)
parity15.to_csv(os.path.join(OUTPUT_DIR,"step1a_parity_diagnostic.csv"),index=False)
display(Markdown(f"**Heston-Lewis parameters**"))
display(heston_param_table)
```

Heston-Lewis parameters

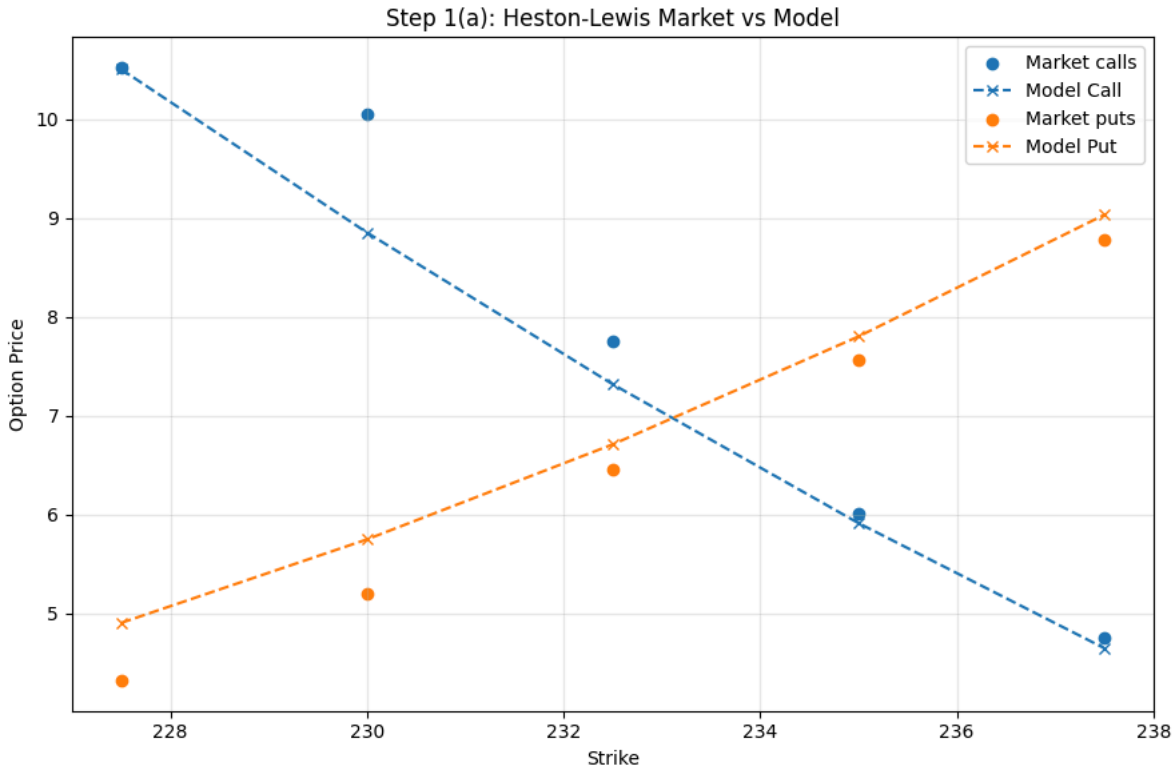
	model	parameter	value
0	Heston-Lewis	kappa	3.022470
1	Heston-Lewis	theta	0.319670
2	Heston-Lewis	sigma_v	1.316672
3	Heston-Lewis	rho	-0.999000
4	Heston-Lewis	v0	0.081105
5	Heston-Lewis	MSE	0.246449
6	Heston-Lewis	RMSE	0.496436

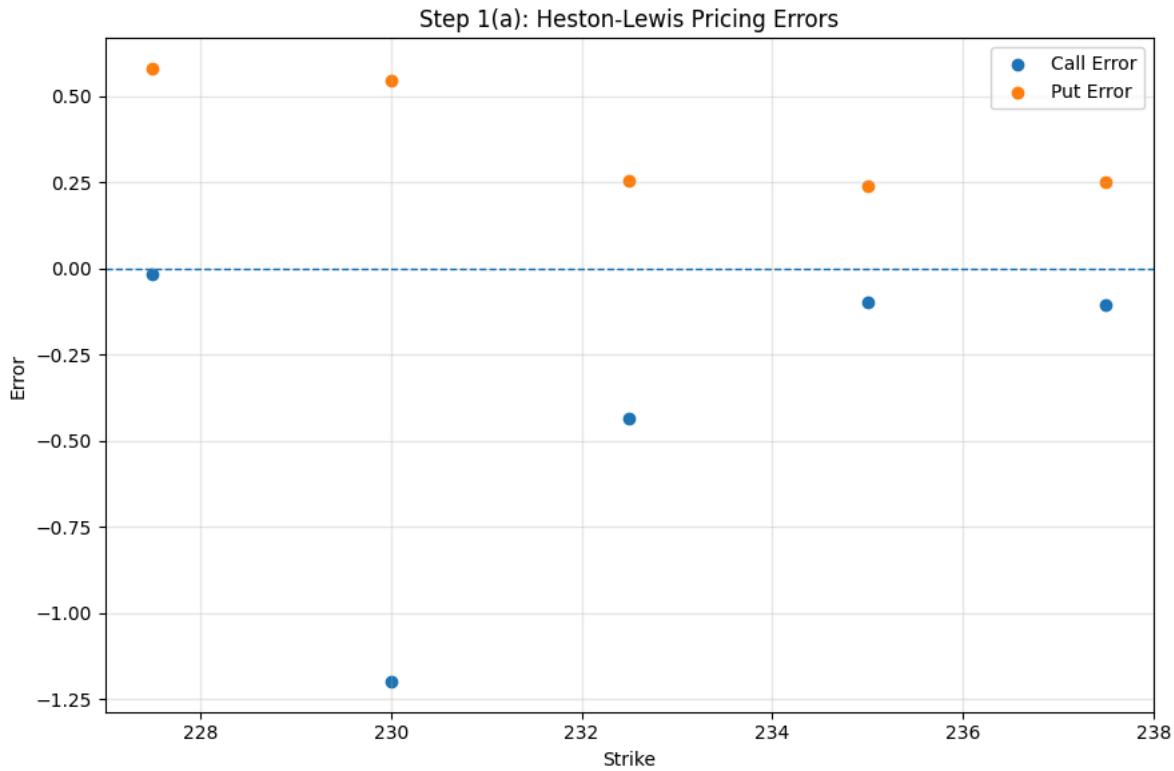
Step 1(a) - fit table and plots

```
heston_model_calls = heston_lewis_call_prices(K15,T15,heston_params)
heston_fit = make_fit_table(K15,C15,P15,heston_model_calls,T15)
heston_fit.to_csv(os.path.join(OUTPUT_DIR,"step1a_fit_table.csv"),index=False)

display(Markdown(f"**Heston-Lewis fit table**"))
display(heston_fit)

plot_fit(heston_fit,"Step 1(a): Heston-Lewis Market vs Model","step1a_heston_lewis_fit.png")
plot_errors(heston_fit,"Step 1(a): Heston-Lewis Pricing Errors","step1a_heston_lewis_errors.png")
```

Heston-Lewis fit table

	Strike	Market Call	Model Call	Call Error	Market Put	Model Put	Put Error
0	227.5	10.52	10.505066	-0.014934	4.32	4.900408	0.580408
1	230.0	10.05	8.852315	-1.197685	5.20	5.745408	0.545408
2	232.5	7.75	7.316563	-0.433437	6.45	6.707407	0.257407
3	235.0	6.01	5.910971	-0.099029	7.56	7.799566	0.239566
4	237.5	4.75	4.644812	-0.105188	8.78	9.031158	0.251158






```
'/content/output_dir/step1a_heston_lewis_errors.png'
```

Start coding or generate with AI.

Bates-Carr-Madan Pricing Model

```python
@dataclass
class BatesParams:
    kappa: float
    theta: float
    sigma_v: float
    rho: float
    v0: float
    lambda_j: float
    mu_j: float
    delta_j: float


def heston_log_stock_cf(u, T, r, p: HestonParams):
    return np.exp(1j * u * np.log(S0)) * heston_return_cf(u, T, r, p)


def bates_log_stock_cf(u, T, r, p: BatesParams):
    hp = HestonParams(p.kappa, p.theta, p.sigma_v, p.rho, p.v0)
    comp = np.exp(p.mu_j + 0.5 * p.delta_j**2) - 1.0
    heston_part = heston_log_stock_cf(u, T, r - p.lambda_j * comp, hp)
    jump_part = np.exp(p.lambda_j * T * (np.exp(1j * u * p.mu_j - 0.5 * p.delta_j**2 * u**2) - 1.0))

    return heston_part * jump_part


def carr_madan_call_prices(strikes, T, cf_func, alpha=1.5):
    u = U_CM
    k = np.log(strikes)
    # cf = cf_func(u - 1j * (alpha + 1.0))
    cf = np.asarray(cf_func(u - 1j * (alpha + 1.0))).reshape(-1)
    denom = alpha**2 + alpha - u**2 + 1j * (2.0 * alpha + 1.0) * u
    integrand = np.real(
        np.exp(-1j * u[:, None] * k[None, :]) *
        (np.exp(-RISK_FREE_RATE * T) * (cf / denom)[:, None])

    )


    integral = W_CM @ integrand
    return np.maximum(np.exp(-alpha * k) * integral / np.pi, 0.0)


def bates_carr_madan_call_prices(strikes, T, p: BatesParams):
    return carr_madan_call_prices(
        strikes,
        T,
        lambda u: bates_log_stock_cf(u, T, RISK_FREE_RATE, p),
        alpha=1.5
    )


def bates_residuals(x, K, MC, MP, T):
    p = BatesParams(*x)

    try:
        with np.errstate(over="ignore", invalid="ignore", divide="ignore"):
            c = bates_carr_madan_call_prices(K, T, p)
            pu = calls_to_puts_by_parity(c, K, T)
            r = np.r_[c - MC, pu - MP]

        return r if np.all(np.isfinite(r)) else np.ones(2 * len(K)) * 1e5

    except Exception:
        return np.ones(2 * len(K)) * 1e5


def bates_mse(x, K, MC, MP, T):
    return float(np.mean(bates_residuals(x, K, MC, MP, T)**2))


def calibrate_bates_carr_madan(K, MC, MP, T):
    bounds = [
        (0.01, 20.0),
        (1e-6, 2.0),
        (0.01, 5.0),
        (-0.999, 0.999),
        (1e-6, 2.0),
        (0.0, 5.0),
        (-0.5, 0.5),
        (1e-6, 1.5)
    ]
```

```
    obj = lambda x: bates_mse(x, K, MC, MP, T)

    g = differential_evolution(
        obj,
        bounds,
        seed=888,
        maxiter=120,
        popsize=12,
        tol=1e-8,
        polish=False,
        updating="immediate",
        workers=1
    )

    l = minimize(
        obj,
        g.x,
        method="L-BFGS-B",
        bounds=bounds,
        options={"maxiter": 2000, "ftol": 1e-14, "gtol": 1e-8}
    )

    return BatesParams(*l.x), obj(l.x)
```

Step 2(b) - slice 60 day options & paraity check

```
K60,C60,P60,T60 =get_maturity_slice(option_data,60)
parity60,party60_lb=parity_diagnostic(K60,C60,P60,T60)

display(Markdown(f"**60-day slice:** {len(K60)} strikes,T= {T60:.5f} years"))
display(parity60)

display(Markdown(f"Parity-constrained lower bounds MSE: **{party60_lb:0.6f}**"))
```

60-day slice: 5 strikes,T= 0.24000 years

	strike	call_minus_put	S0_minus_discounted_K	parity_deviation
0	227.5	5.75	6.217528	-0.467528
1	230.0	5.50	3.726511	1.773489
2	232.5	3.49	1.235495	2.254505
3	235.0	1.30	-1.255521	2.555521
4	237.5	-0.52	-3.746537	3.226537

Parity-constrained lower bounds MSE: **1.269393**

Step 2(b) - calibrate Bates-Carr-Madan & report parameters

```
bates_params,bates_mse_val = calibrate_bates_carr_madan(K60,C60,P60,T60)
bates_param_table = params_to_dataframe(bates_params,bates_mse_val,"Bates-Carr-Madan",["kappa","theta","sigma_v","rho","v0'

bates_param_table.to_csv(os.path.join(OUTPUT_DIR,"step2b_bates_carr_madan_params.csv"),index=False)
parity60.to_csv(os.path.join(OUTPUT_DIR,"step2_parity_diagnostics.csv"),index=False)
display(Markdown(f"**Bates-Carr-Madan parameters**"))
display(bates_param_table)
```

Bates-Carr-Madan parameters

	model	parameter	value
0	Bates-Carr-Madan	kappa	0.010000
1	Bates-Carr-Madan	theta	0.000001
2	Bates-Carr-Madan	sigma_v	4.382365
3	Bates-Carr-Madan	rho	-0.999000
4	Bates-Carr-Madan	v0	0.026626
5	Bates-Carr-Madan	lambda_j	1.901816
6	Bates-Carr-Madan	mu_j	0.207446
7	Bates-Carr-Madan	delta_j	0.026248
8	Bates-Carr-Madan	MSE	1.326750
9	Bates-Carr-Madan	RMSE	1.151847

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Step 2(b) - fit table and plots

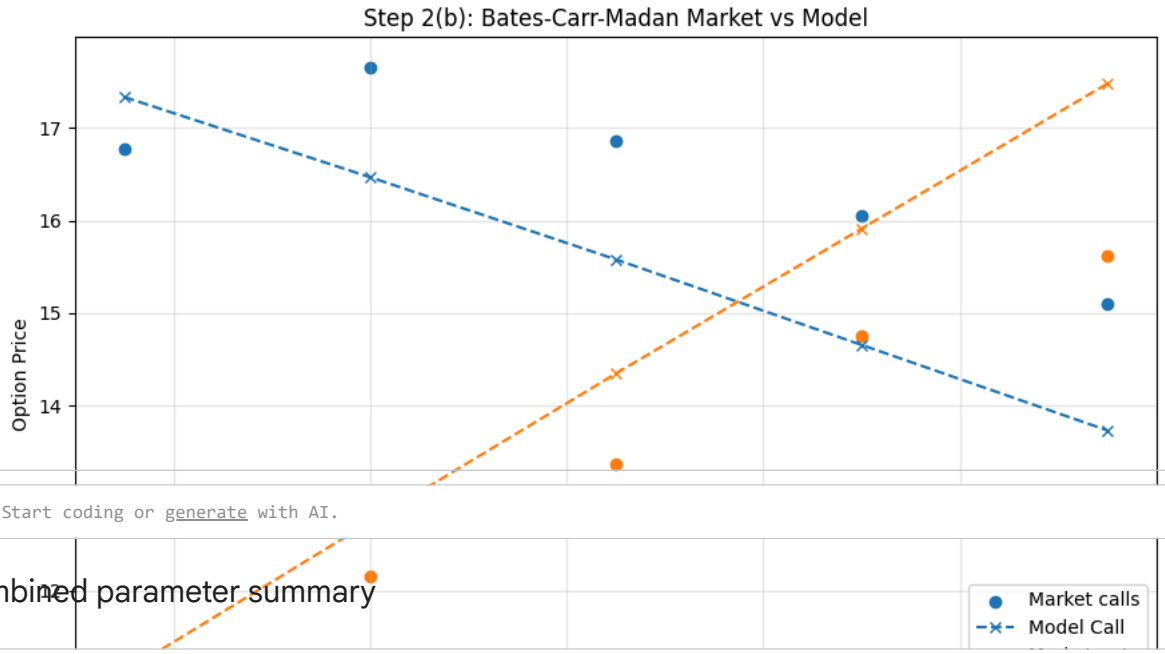
```
bates_model_calls= bates_carr_madan_call_prices(K60,T60,bates_params)
bates_fit = make_fit_table(K60,C60,P60,bates_model_calls,T60)
bates_fit.to_csv(os.path.join(OUTPUT_DIR,"step2b_bates_carr_madan_fit_table.csv"),index=False)

display(Markdown(f"**Bates-Carr-Madan fit table (60-day)**"))
display(bates_fit)

plot_fit(bates_fit,"Step 2(b): Bates-Carr-Madan Market vs Model","step2b_bates_carr_madan_fit.png")
plot_errors(bates_fit,"Step 2(b): Bates-Carr-Madan Pricing Errors","step2b_bates_carr_madan_errors.png")
```

Bates-Carr-Madan fit table (60-day)

	Strike	Market Call	Model Call	Call Error	Market Put	Model Put	Put Error
0	227.5	16.78	17.334004	0.554004	11.03	11.116476	0.086476
1	230.0	17.65	16.468066	-1.181934	12.15	12.741554	0.591554
2	232.5	16.86	15.579209	-1.280791	13.37	14.343714	0.973714
3	235.0	16.05	14.654879	-1.395121	14.75	15.910400	1.160400
4	237.5	15.10	13.731165	-1.368835	15.62	17.477702	1.857702



Start coding or generate with AI.

combined parameter summary

```
summary_df = pd.concat([heston_param_table,bates_param_table],ignore_index=True)
summary_df

display(Markdown(f"**Combined parameter summary**"))
display(summary_df)
```

Combined parameter summary

model	parameter	value